

Club Filter without Choice

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Global Assumption 1. Suppose that

1. ZF holds,
2. S is an infinite set,
3. DC_S holds. □

Lemma 2. Set $[S]^\omega$ is non-empty. □

Definition 3. Let $\mathcal{F}$ be a filter on $[S]^\omega$.

- a. $\mathcal{F}$ is said to be **fine** iff for all $\tau \in [S]^\omega$,

$$\{\sigma \in [S]^\omega : \tau \subseteq \sigma\} \in \mathcal{F}.$$

- b. $\mathcal{F}$ is said to be **normal** iff for all A which are $\mathcal{F}$ positive, for all

$$F : A \rightarrow [S]^\omega$$

satisfying that

$$\forall \sigma \in A, \emptyset \subset F(\sigma) \subseteq \sigma,$$

there exists $a \in S$ such that

$$\{\sigma \in A : a \in F(\sigma)\}$$

is $\mathcal{F}$ -positive.

- c. $\mathcal{F}$ is said to have the **diagonal intersection property** iff for all families $(A_a : a \in S)$ of sets in $\mathcal{F}$, the set

$$\Delta_{a \in S} A_a := \{\sigma \in [S]^\omega : \forall a \in \sigma, \sigma \in A_a\}$$

belongs to $\mathcal{F}$. □

Proposition 4. Let $\mathcal{F}$ be a filter on $[S]^\omega$. Then $\mathcal{F}$ is normal iff $\mathcal{F}$ has the diagonal intersection property.

Proof. By the proof of [Sol78, Lemma 4.5]. □

Lemma 5. *Suppose that $\mathcal{F}$ is a normal, fine filter on $[S]^\omega$. Then $\mathcal{F}$ is ω_1 -complete.*

Proof. Let $(A_n : n < \omega) \in \mathcal{F}^\omega$ be arbitrary. There exists an injection $(a_n : n < \omega) \in S^\omega$. We define $(B_a : a \in S) \in \mathcal{F}^S$ as follows: for $a \in S$,

$$B_a := \begin{cases} A_n, & n < \omega, a = a_n \\ S, & \text{otherwise.} \end{cases}$$

Since $\mathcal{F}$ is fine, we have that

$$C := \{\sigma \in [S]^\omega : \{a_n\}_{n < \omega} \subseteq \sigma\} \in \mathcal{F}.$$

Note that

$$\bigcap_{n < \omega} A_n \cap C = \bigtriangleup_{a \in S} B_a \cap C.$$

Since $\mathcal{F}$ is normal, it follows that $\bigcap_{n < \omega} A_n \in \mathcal{F}$. $\square$

Definition 6. Let $A \subseteq S$. The game $\mathfrak{D}^{\text{club}}(S, A)$ is a length ω two player game of the form

$$\begin{array}{c|ccc} \text{I} & a_0 & a_1 & \cdots \\ \hline \text{II} & b_0 & b_1 & \cdots \end{array}$$

where for all $n < \omega$, $a_n, b_n \in S$. Player II wins iff $\{a_n, b_n : n < \omega\} \in A$. $\square$

Notation 7. We define μ_S to consist of all $A \subseteq [S]^\omega$ for which Player II has a winning quasi-strategy in $\mathfrak{D}^{\text{club}}(S, A)$.

Proposition 8. μ_S is a fine filter.

Proof. To see that μ_S is a filter, the only non-trivial part is to verify the closure for intersections. This is argued similarly to the countable completeness of [Sol78, Lemma 4.1], but we work with quasi-strategies instead of strategies. Our case is simpler inasmuch that we are only considering intersections of two sets. Let us explain why μ_S is fine. Let $\sigma \in [S]^\omega$ and let

$$A := \{\sigma' \in [S]^\omega : \sigma \subseteq \sigma'\}.$$

We need to give a winning quasi-strategy for Player II in $\mathfrak{D}^{\text{club}}(S, A)$. If we fix an enumeration $(a_n : n < \omega)$ of σ , then Player II can win $\mathfrak{D}^{\text{club}}(S, A)$ by playing a_n 's, regardless of the moves of Player I. $\square$

Proposition 9. *Suppose that $\text{DC}_{\mathcal{P}(S^{<\omega})}$ holds. Then filter μ_S is normal.*

Proof. Suppose that $(A_a : a \in S)$ is a family of sets in μ_S . Implicit in the proof of [Sol78, Lemma 4.6] is a quasi-strategy τ for Player II in the game in the game

$$\mathfrak{D}^{\text{club}}(S, \bigtriangleup_{a \in S} A_a).$$

Note that we do not pick a family $(\tau_a : a \in S)$ of winning quasi-strategy for Player II in the games

$$(\mathfrak{D}^{\text{club}}(S, A_a) : a \in A)$$

as it is done there, but we play a tree of auxiliary games according to all possible such quasi-strategies. In the end, if f is a play according to τ , we can use $\text{DC}_{\mathcal{P}(S^{<\omega})}$ to be pick a sequences of $(\tau_a : a \in \text{ran}(f))$ of winning quasi-strategies for Player II in the games

$$(\mathfrak{D}^{\text{club}}(S, A_a) : a \in \text{ran}(f))$$

which ensure that for all $a \in \text{ran}(f)$, $\text{ran}(f) \in A_a$. $\square$

Definition 10. A **supercompactness measure** on $[S]^\omega$ is a normal, fine ultrafilter on $[S]^\omega$. $\square$

Proposition 11. Suppose that μ is a supercompactness measure on $[S]^\omega$. Then $\mu_S \subseteq \mu$.

Proof.

- 1° Let $A \in \mu_S$ be arbitrary. We want to show that $A \in \mu$.
- 2° Let us assume otherwise. Then $B := [S]^\omega - A \in \mu$.
- 3° Let τ be a winning quasi-strategy for Player II in $\mathfrak{D}^{\text{club}}(S, A)$ and let G be the game tree for a game of the form

$$\begin{array}{c|ccc} \text{I} & a_0 & a_1 & \cdots \\ \hline \text{II} & s_0 & s_1 & \cdots \end{array}$$

where for all $n < \omega$, $a_n \in S$ and $s_n \in \tau$. For $p \in G$, we denote a_n^p and s_n^p the corresponding a_n 's and s_n 's.

- 4° For $\sigma \in B$, the game $\mathfrak{D}^{\text{aux}}(\sigma)$ is played along G with the following conditions:
 - a. Player I must ensure that for all $n < \omega$, $a_n \in \sigma$,
 - b. Player II must ensure that for all $n < \omega$, $s_n \in \tau$, $s_n \trianglelefteq s_{n+1}$, and $a_n \in \text{ran}(s_n) \subseteq \sigma$.
 - c. The first player to break one of these rules loses.
 - d. Infinite plays with no rules broken are won by Player II.

Let us also denote by $R_\sigma \subseteq G$ the rule tree for $\mathfrak{D}^{\text{aux}}(\sigma)$.

- 5° **Claim.** Let $\sigma \in B$. Then Player II does not have a winning quasi-strategy in $\mathfrak{D}^{\text{aux}}(\sigma)$.

Proof.

- 1' Let us assume otherwise and let ϕ be one such quasi-strategy.
- 2' By DC_S , there exists an infinite play f of $\mathcal{D}^{\text{aux}}(\sigma)$ according to ϕ in which Player I plays some enumeration of σ .
- 3' Since f is according to ϕ , it is won by Player II. Letting $f = (a_n, s_n : n < \omega)$ and letting s be the union of $(s_n)_n$, the rules imply that

$$\sigma = \{a_n\}_n \subseteq \text{ran}(s) \subseteq \sigma.$$

- 4' Thus, $\sigma = \text{ran}(s)$ and s is a play of $\mathcal{D}^{\text{club}}(S, A)$ according to τ , so $\sigma \in A$. This is a contradiction.

□

- 6° By Gale-Stewart theorem, for all $\sigma \in B$, there exists a winning quasi-strategy for Player I in $\mathcal{D}^{\text{aux}}(\sigma)$. Note that any such quasi-strategy is a subtree of G .
- 7° **Claim.** There exists a family $(\phi_\sigma : \sigma \in B)$ such that for all $\sigma \in B$, ϕ_σ is a winning quasi-strategy for Player I in $\mathcal{D}^{\text{aux}}(\sigma)$.

Proof.

- 1' For a given $\sigma \in B$, we will define a winning quasi-strategy ϕ_σ for Player I in $\mathcal{D}^{\text{aux}}(\sigma)$.
- 2' Let Q_σ consist of all winning quasi-strategies for Player I in $\mathcal{D}^{\text{aux}}(\sigma)$. Note that $Q_\sigma \neq \emptyset$.
- 3' For $\phi \in Q_\sigma$, let $\bar{\phi} := \phi \cap R_\sigma$. Note that $\bar{\phi}$ is a well-founded tree.
- 4' For $\phi \in Q_\sigma$ and $p \in \bar{\phi}$, let us define $\rho_{\phi, p}$ to be the rank of p in $\bar{\phi}$.
- 5' For $p \in G$, let us define

$$\rho_p := \min\{\rho_{\phi, p} : \phi \in Q_\sigma, p \in \bar{\phi}\},$$

where $\min \emptyset := \infty$ is a formal object greater than all the ordinals.

- 6' Let $\bar{\phi}_\sigma$ be defined recursively as follows:

- $\emptyset \in \bar{\phi}_\sigma$
- for all $p \in \bar{\phi}_\sigma$ of odd length, for all s such that $p^\frown(s) \in R_\sigma$, we set $p^\frown(s) \in \bar{\phi}_\sigma$
- for all $p \in \bar{\phi}_\sigma$ of even length, for all $a \in S$ such that $\rho_{p^\frown a} < \rho_p$, we set $p^\frown a \in \bar{\phi}_\sigma$.

Note that the function $\rho_{(-)}$ decreases along $\bar{\phi}_\sigma$, which implies that $\bar{\phi}_\sigma$ is a well-founded tree.

- 7' Let ϕ_σ consist of all $p \in G$ which extend some leaf of $\bar{\phi}_\sigma$. We want to show that ϕ_σ is a winning quasi-strategy for Player I in $\mathcal{D}^{\text{aux}}(\sigma)$.

- 8' Observe that for all $p \in \bar{\phi}_\sigma$, $\rho_p < \infty$. This means that for all $p \in \bar{\phi}_\sigma$, there exists $\phi \in Q_\sigma$ such that $\rho_p = \rho_{p,\phi}$.
- 9' For all $p \in \bar{\phi}_\sigma$ and for all $\phi \in Q_\sigma$ such that $\rho_p = \rho_{p,\phi}$, every one point extension of p that is in $\bar{\phi}$ is also in $\bar{\phi}_\sigma$. This is sufficient for the conclusion that ϕ_σ is a quasi-strategy for Player I.
- 10' The fact that $\bar{\phi}_\sigma$ is well-founded implies that ϕ_σ is winning.

□

8° Let T consist of $p \in G$ such that for μ -almost all σ , $p \in \phi_\sigma \cap R_\sigma$. Note that T is a tree and $\emptyset \in T$.

9° **Claim.** T does not have a maximal element.

Proof.

1' Let $p \in T$ be arbitrary. We have that

$$B_p := \{\sigma \in B : p \in \phi_\sigma \cap R_\sigma\} \in \mu.$$

2' Suppose first that p is of odd length. Let us fix $\sigma \in B_p$.

3' Note that there exists s such that $p^\frown(s) \in R_\sigma$. Since ϕ_σ is a quasi-strategy for Player I, we get that $p^\frown(s) \in \phi_\sigma$.

4' Since

$$\{\sigma' \in B : p^\frown(s) \in \phi_{\sigma'} \cap R_{\sigma'}\} \supseteq \{\sigma' \in B : \sigma \subseteq \sigma'\} \in \mu,$$

we see that $p^\frown(s) \in T$.

5' Let us now suppose that p is of even length. Then for all $\sigma \in B_p$, there exists $a \in S$ such that $p^\frown a \in \phi_\sigma \cap R_\sigma$. Note that in such a case, the rules ensure that $a \in \sigma$.

6' Let

$$F : B_p \rightarrow [S]^\omega : \sigma \mapsto \{a \in S : p^\frown a \in \phi_\sigma \cap R_\sigma\}.$$

Then for all $\sigma \in B_p$, $\emptyset \subset F(\sigma) \subseteq \sigma$.

7' By the normality of μ , there exists $a \in S$ such that

$$\{\sigma \in B_p : a \in F(\sigma)\} \in \mu.$$

8' This shows that $p^\frown a \in T$.

□

10° Since DC_S holds, there is a branch f through T .

11° Since μ is ω_1 -complete, it follows that

$$\{\sigma \in B : \forall n < \omega, f \upharpoonright n \in \phi_\sigma \cap R_\sigma\} \in \mu.$$

In particular, this set contains at least one σ .

12° Since f is according to ϕ_σ , so it is won by Player I. However, since f is a branch through R_σ , Player II did not break any rule. This is obviously a contradiction. □

Lemma 12. *Suppose that $\text{AD}_{\mathbb{R}}$ holds and that $\mathbb{R}$ surjects onto S . Then in every game on S , one of the players has a winning quasi-strategy.*

Proof.

1° Let $A \subseteq S$ be arbitrary. We want to show that one of the players in the game $\mathcal{O}(S, A)$ on S with A as the payoff set for Player II has a winning quasi-strategy.

2° Let $\pi : \mathbb{R} \twoheadrightarrow S$ be a surjection and let

$$B := \{y \in \mathbb{R}^\omega : \pi \circ y \in A\}$$

. By $\text{AD}_{\mathbb{R}}$, one of the players in the game on $\mathbb{R}$ with B as the payoff set for Player II has a winning strategy. We can assume without loss of generality that it is Player I who has a winning strategy τ .

3° Regarding τ as a tree, let $\sigma := \{\pi \circ y : y \in \tau\}$. Then σ is a quasi-strategy for Player I in $\mathcal{O}(S, A)$. We want to show that it is winning.

4° Let f be a play according to σ . By $\text{DC}_{\mathbb{R}}$, there exists g according to τ such that $f = \pi \circ g$.

5° Since g is according to τ , we have that $g \in B^{\mathbb{G}}$. This implies $f \in A^{\mathbb{G}}$, as required. □

Proposition 13. *Suppose that $\text{AD}_{\mathbb{R}}$ holds and that $\mathbb{R}$ surjects onto S . Then μ_S is a supercompactness measure. Furthermore, it is the only supercompactness measure on $[S]^\omega$.¹*

Proof. By Lemma 12, either Player I or Player II has a winning quasi-strategy in $\mathcal{O}^{\text{club}}(S, A)$. If Player II has a winning quasi-strategy, then $A \in \mu_S$ by the definition. On the other hand, if Player I has a winning quasi-strategy in $\mathcal{O}^{\text{club}}(S, A)$, then we can easily show that Player II has a winning quasi-strategy in $\mathcal{O}^{\text{club}}(S, A^{\mathbb{G}})$, which means that $A^{\mathbb{G}} \in \mu_S$. Thus, μ_S is a fine ultrafilter on $[S]^\omega$. To see that it is also normal, we can run the proof of [Sol78, Lemma 4.6]. On the other hand, the uniqueness follows from 11. □

¹That there exists a unique supercompactness measure on $[S]^\omega$ was proved in [Woo21, Theorem 1.5].

Proposition 14. *Suppose that S' is an infinite set and $\pi : S \rightarrow S'$. Then $\text{DC}_{S'}$ holds and for all $A \subseteq [S']^\omega$, $A \in \mu_{S'}$ if and only if*

$$\{\sigma \in [S]^\omega : \pi[\sigma] \in A\} \in \mu_S.$$

Proof. That $\text{DC}_{S'}$ holds is routinely verified. Let $\pi_* : [S]^\omega \rightarrow [S']^\omega : \sigma \mapsto \pi[\sigma]$. We need to show that $A \in \mu_{S'}$ if and only if $\pi_*^{-1}[A] \in \mu_S$. Observe that under the correspondence

$$S^{<\omega} \rightarrow (S')^{<\omega} : p \mapsto \pi \circ p,$$

quasi-strategies for Player II for games on S correspond to quasi-strategies for Player II for games on S' . Since DC_S holds, the property of a quasi-strategy of being winning is invariant under this correspondence, which ensures the conclusion. $\square$

Proposition 15. *Suppose that*

1. M is an inner model,
2. $S \in M$,
3. $S^\omega \subseteq M$.

Then $\mu_S^M \subseteq \mu_S$. In particular, if μ_S^M is an ultrafilter in M , then $\mu_S^M = \mu_S \cap M$.

Proof. Suppose that $A \in \mu_S^M$. Then there exists τ such that $M \models \text{“}\tau \text{ is a winning quasi-strategy for Player II in } \mathcal{D}^{\text{club}}(S, A)\text{”}$. Since $S^\omega \subseteq M$, we see that τ is a winning quasi-strategy for Player II in $\mathcal{D}^{\text{club}}(S, A)$ (in V). This shows that $A \in \mu_S$. $\square$

Proposition 16. *Suppose that AC holds. Then μ_S is equal to the club filter on $[S]^\omega$.*

Proof. AC allows us to work with strategies instead of quasi-strategies. The conclusion then follows from [Vää11, Proposition 6.19]. $\square$

References

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